

TIA Portal User Manual

Remote access to TIA Portal software

1. In the browser, go to remote.jamk.fi
2. Log in with your student ID
3. On the top left, select the Desktops > All Desktops
4. You should be able to see the remote desktops for which you have received permissions
5. Open a remote desktop called Technology Desktop (click the monitor icon). If you don't see this desktop, contact the teacher.
6. The first time you boot is slower than usual because the system creates a profile for you. In the future, the environment will be slightly faster.
7. On the remote desktop that opens, you will find shortcuts related to TIA Portal. Of these, you will need two:
 1. TIA PORTAL V16
 2. S7-PLCSIM V16 - This is what you use to simulate a ready-made application made with TIA Portal software.

Remember to always save files to H drive! Users' profiles are automatically deleted, causing files stored elsewhere to disappear. (If you are using Remote connection, save file to the C drive and save the file on the H drive only after you finish)

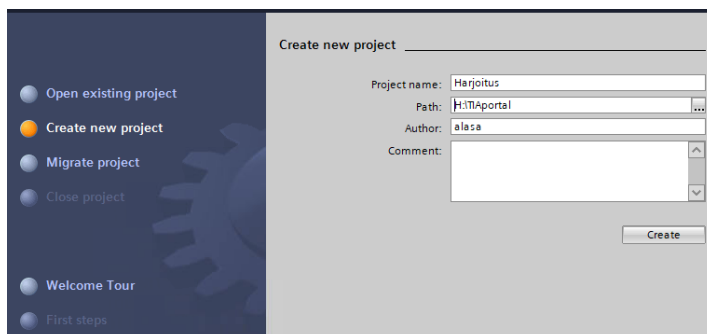
At the end of each work, log out (disconnect) both from the remote desktop and remotely (the cog > logoff in the upper right corner). Do not leave the connection open if you are not using it.

Create a project

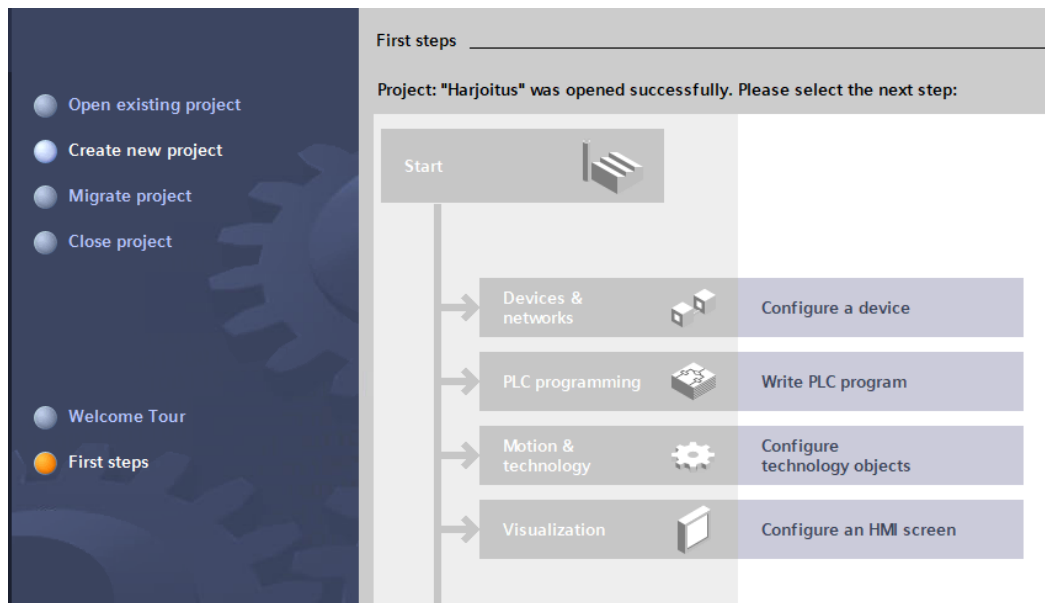
1. From your desktop, open the TIA Portal V16 program



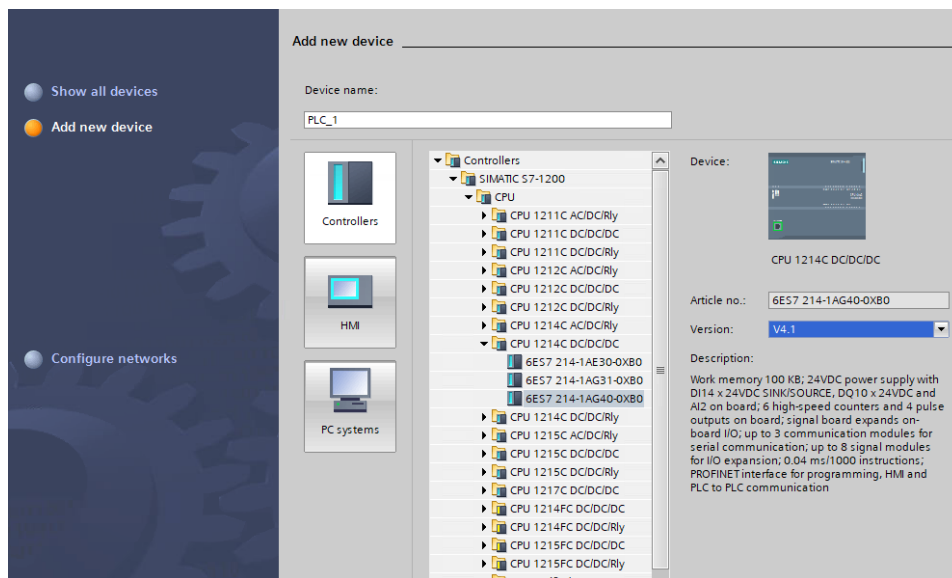
2. Click **Create new** project, name the project and select a save folder. It is worth taking advantage of the H-drive as a storage folder for the exercises. You can also get to the H-station from the school's workstations. Press **Create** on the right side of the view.



3. Select **Configure a device**



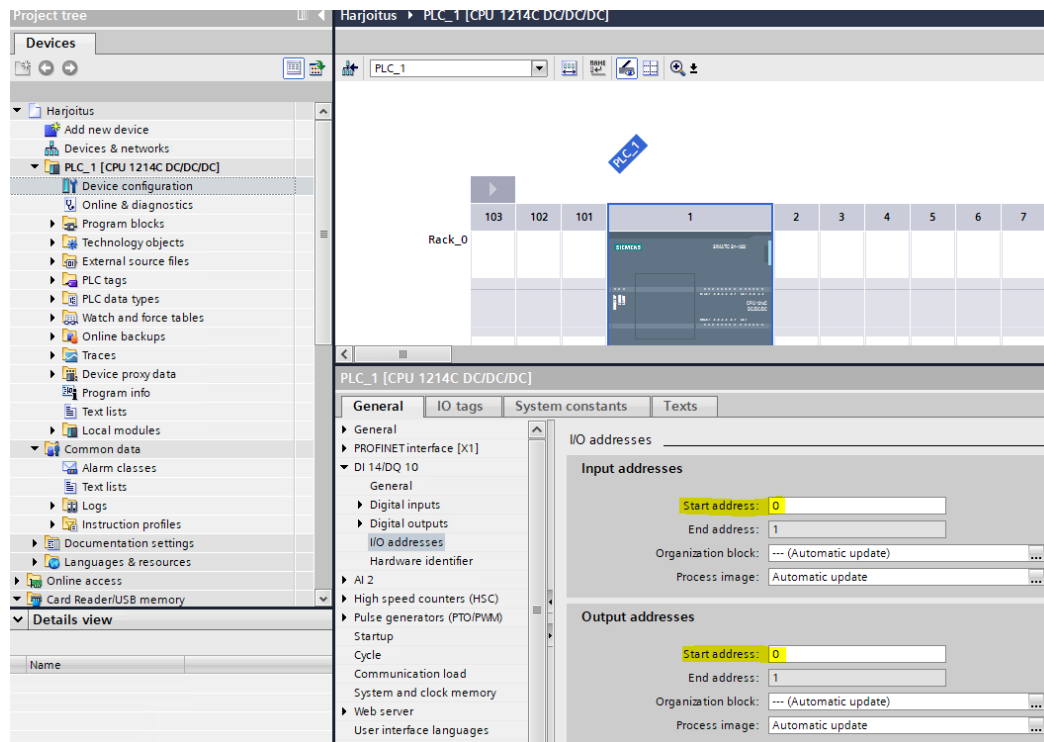
4. Open the **Add new** device view and select the device to use in the project. The laboratory mainly uses CPU 1214C-DC/DC/DC logics. The model can be checked from the side of the logic. Select the device as active, after which you will be able to select the version number (Firmware). If the logic has a different firmware than the program being downloaded, the program will give an error message. The logic is interchangeable at a later stage. At the bottom right, press the **Add** button.



Modification of the I/O specification

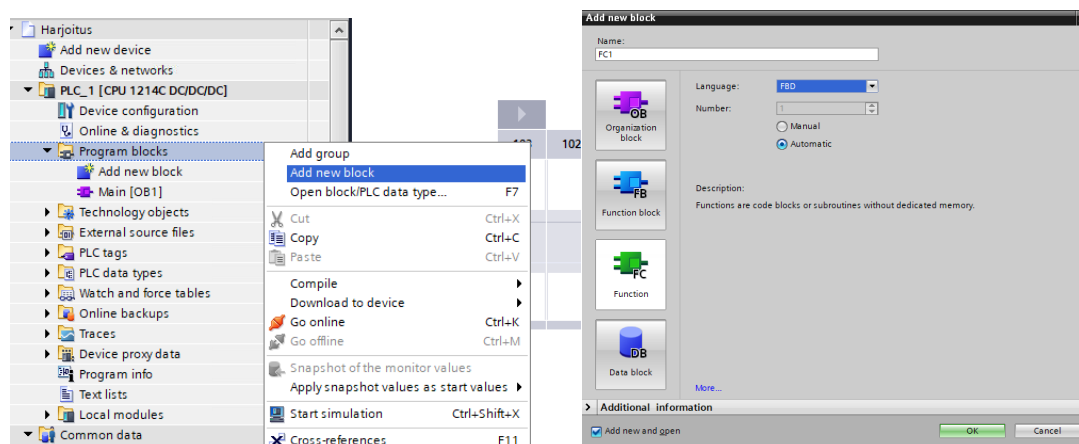
5. From the **Project tree** view on the left, double-tap **Device configuration**. After that, the properties menu will open at the bottom. Select **DI xx/DQ xx I/O addresses** →→

Check/change Input & Output addresses to Start address if necessary.

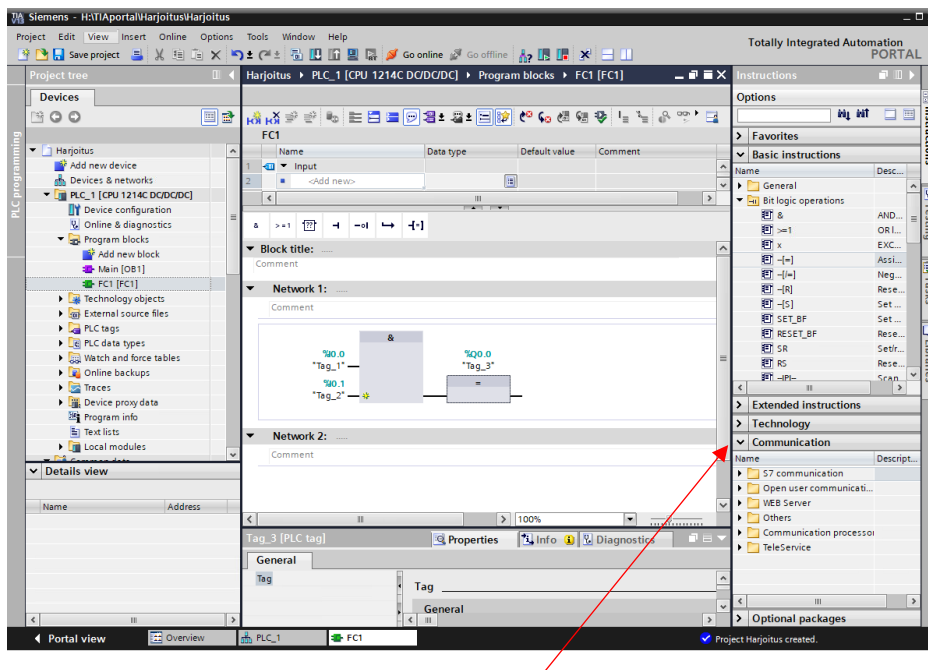


Add a program

6. Select **Program blocks** with the right mouse button and select **Add new block**. Select the action you want to add and the language (Function, FBD) to use. Press **OK**.

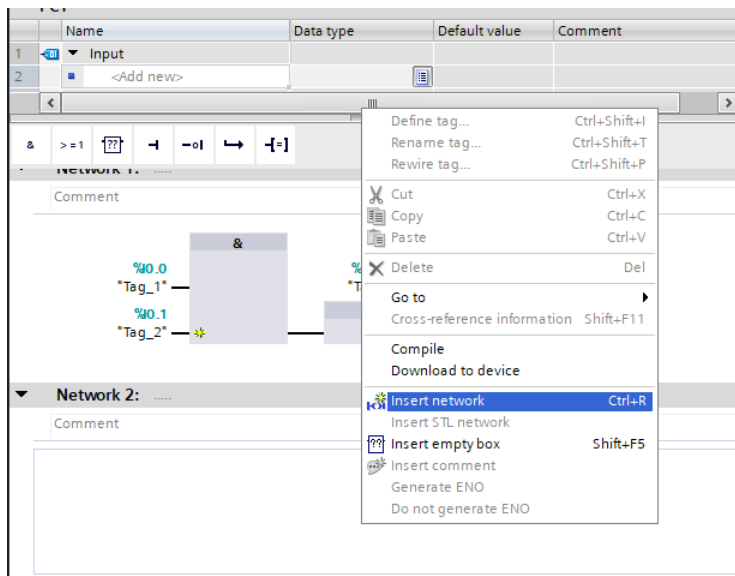


7. Drag the necessary actions from the folders on the right to the Network windows.



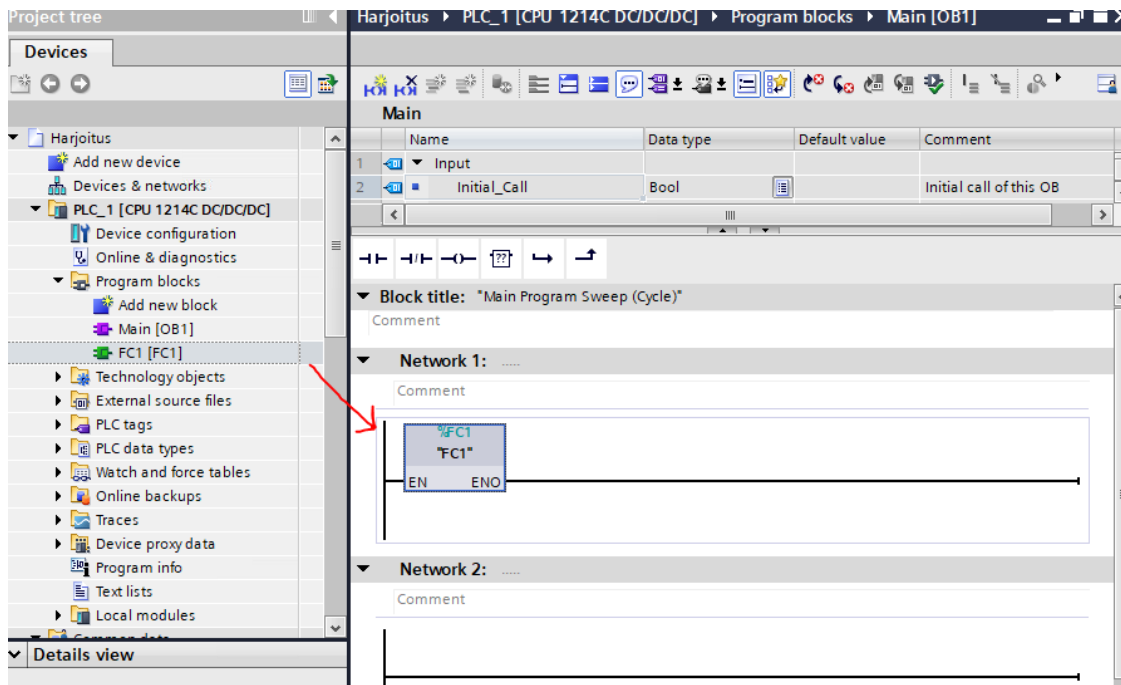
It is worth reducing the extra windows from the arrow.

8. You can add a new Network window by right-clicking on the Network window and selecting **Insert Network**.



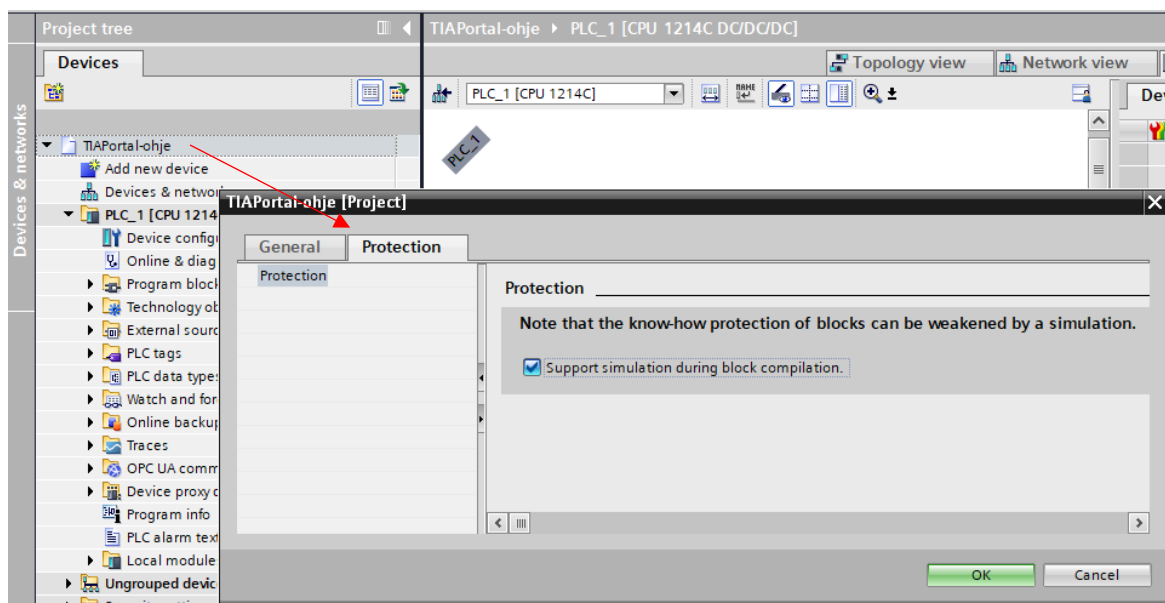
Add a function to a cycle

10. From the Project tree menu, **open Program blocks Main[OB1]** → and drag the block you made into the **Network** window.

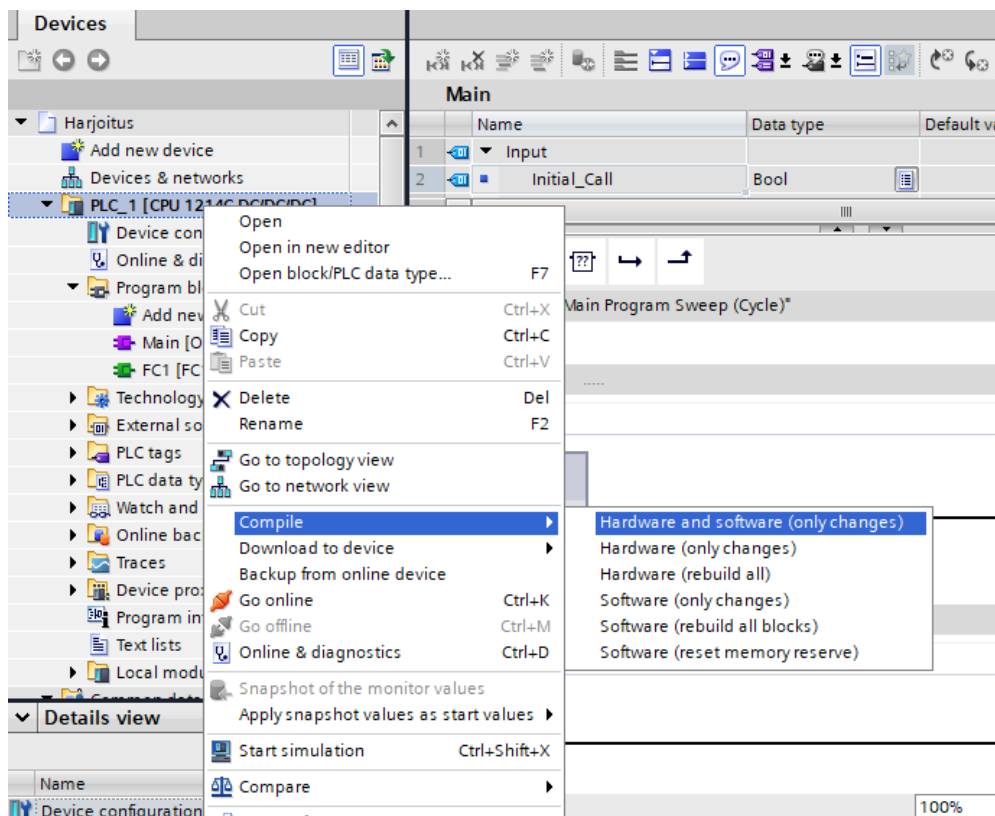


Inspection and download of the program

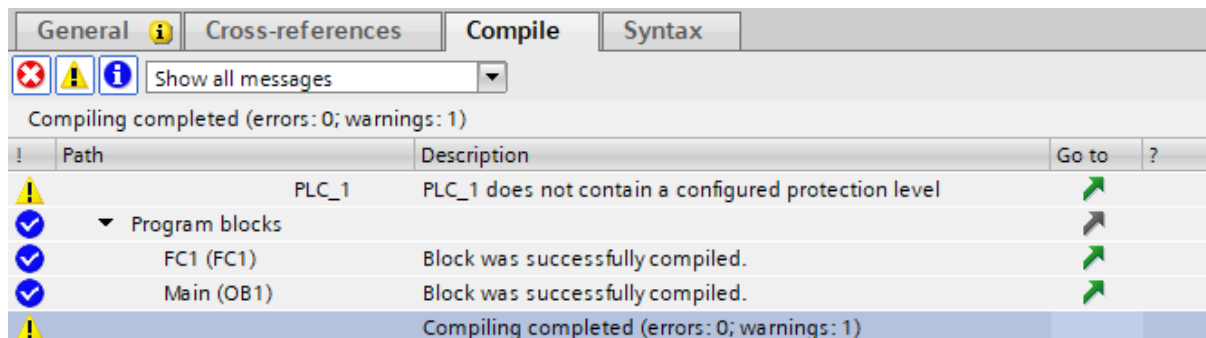
When simulating a program, you need to make sure that simulation is allowed. Select the project with the right mouse and open the **Properties** window. In the window that opens, on the **Protection** tab, set the **Support simulation during block compilation** function to active. Finally, press **OK**.



11. From the Project tree menu, open **PLC** with the right mouse button and select **Compile** → **Hardware and software**

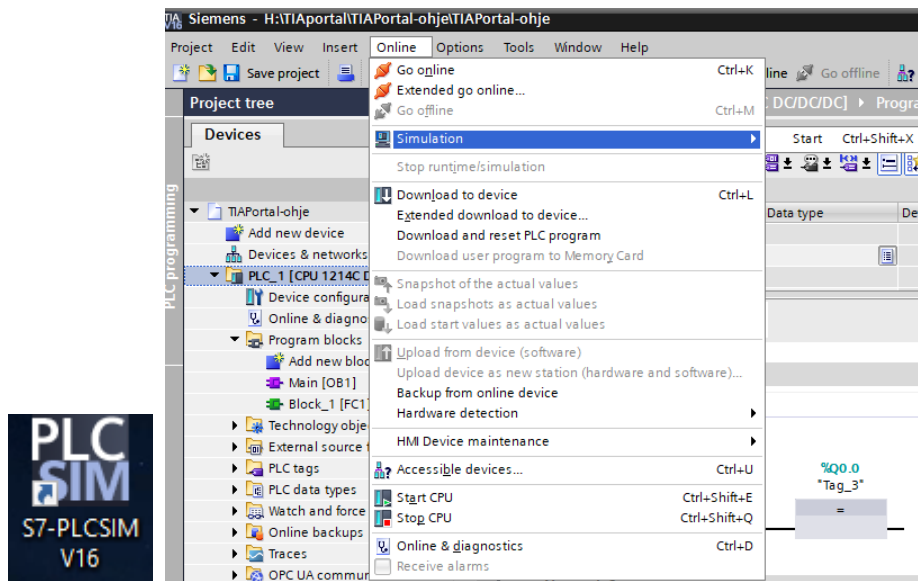


12. At the bottom, check that everything goes through (no errors)

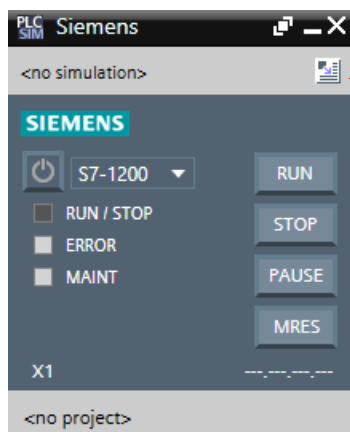


Starting the simulator

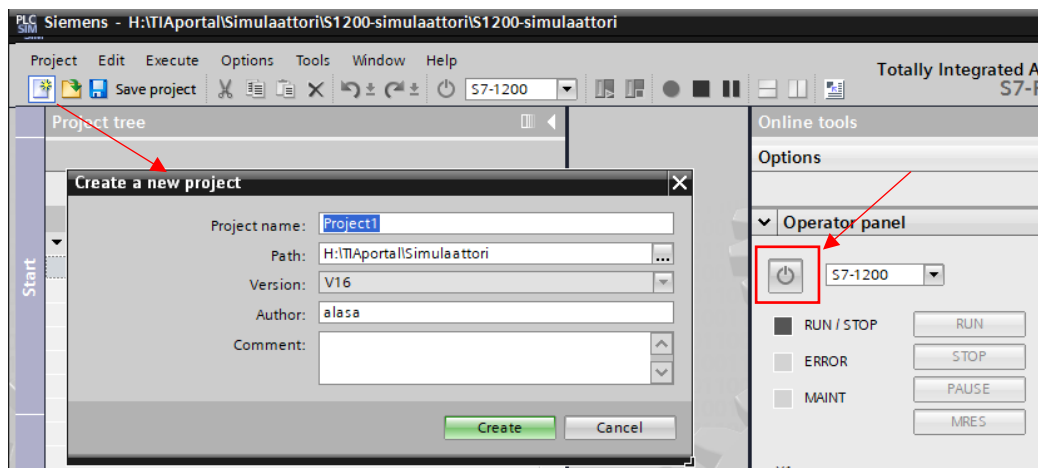
Open the S7-PLCSIM program from the desktop or through TIA Portal Online → Simulation → Start:.



At the top right, click **Switch to Project view**.



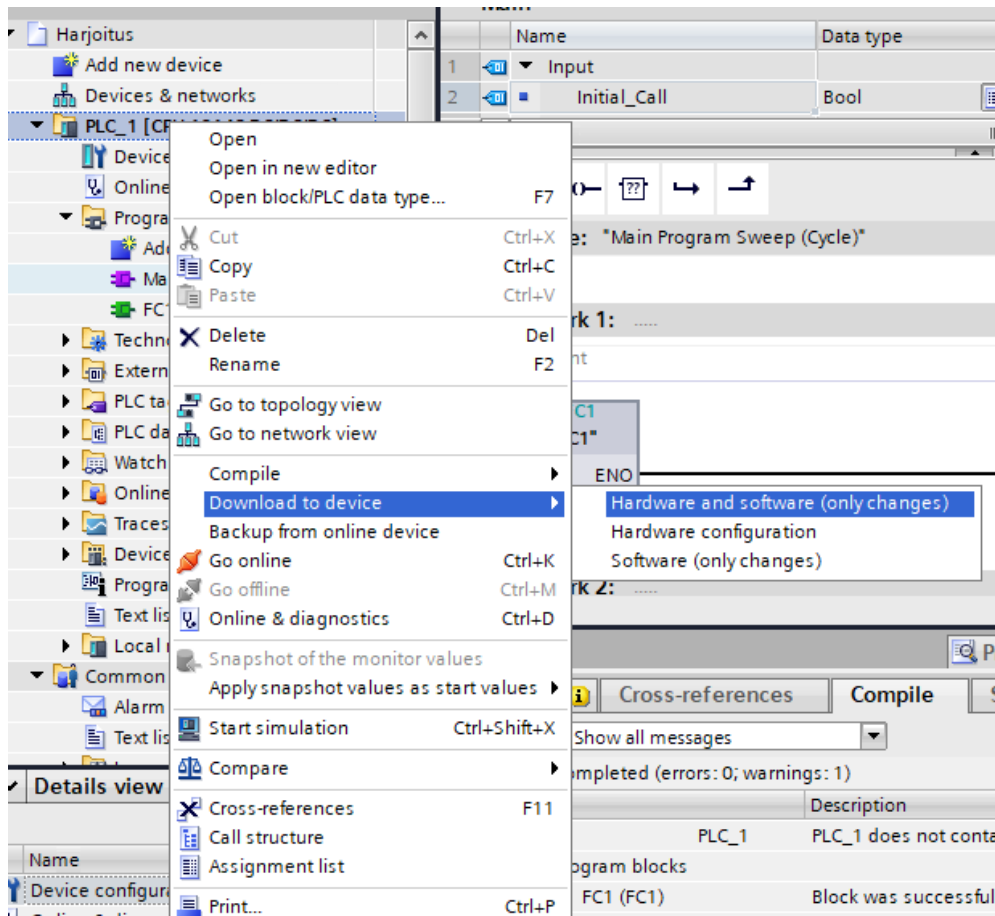
In the upper-left corner of the program, click **New Project**. Name the project and select a save folder. Press **Create**. Launch the simulator from the start button on the right side.



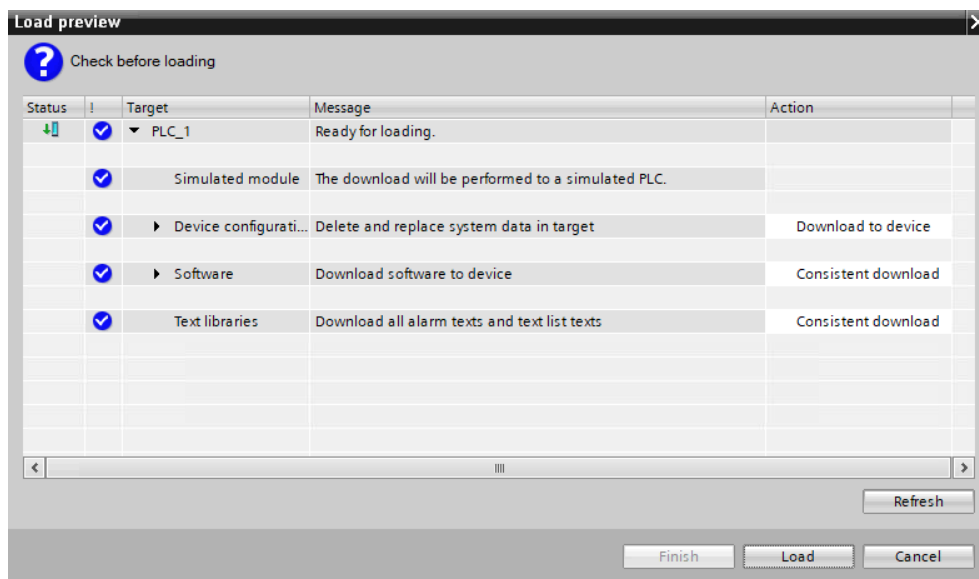
While the simulator is running, the project can be downloaded from the TIA Portal.

Download a program

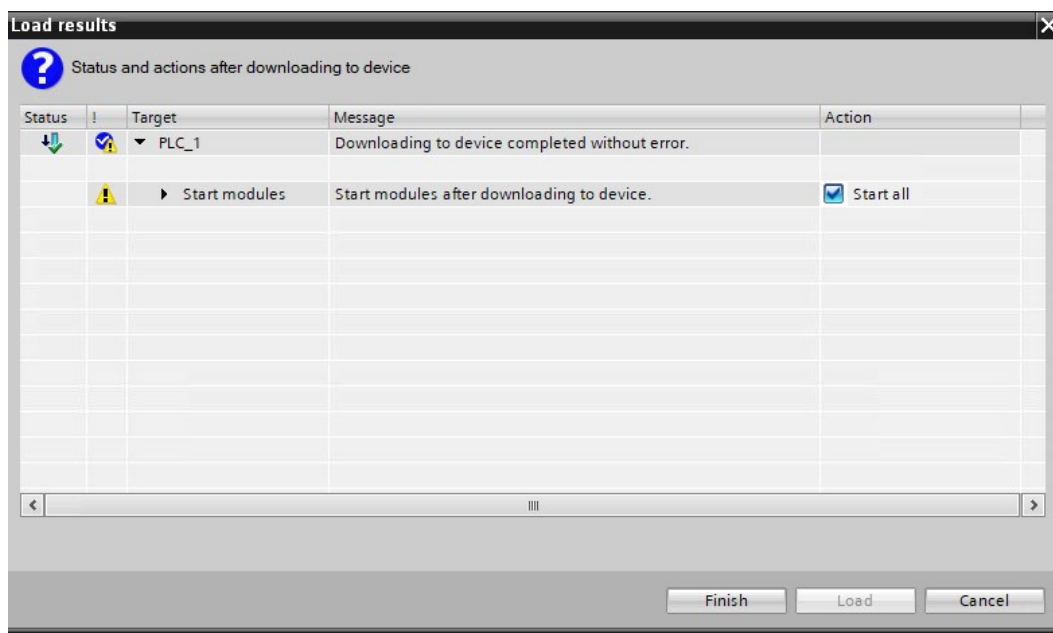
In TIA Portal, from the Project tree menu, open **PLC** with the right mouse button and select **Download to device → Hardware and software**



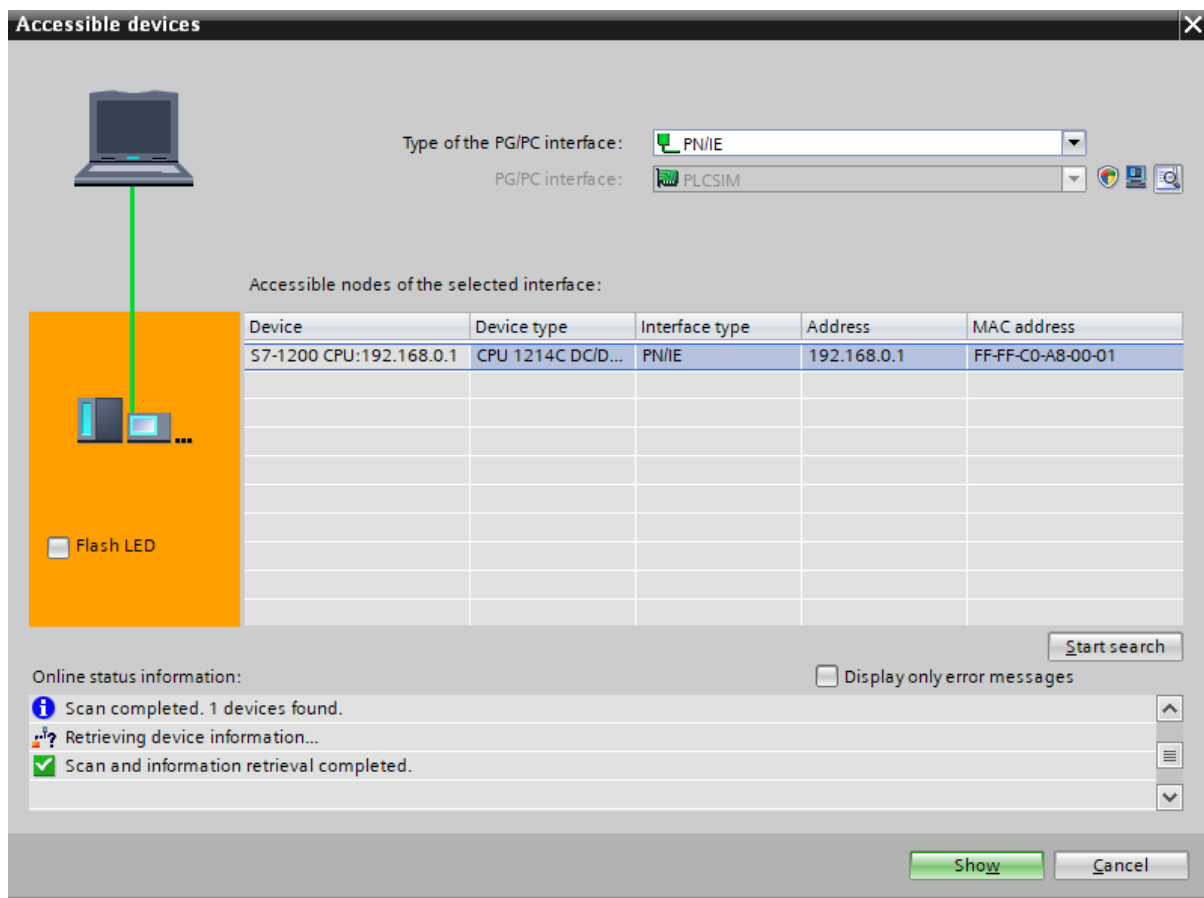
In the window that opens, select **Load**.



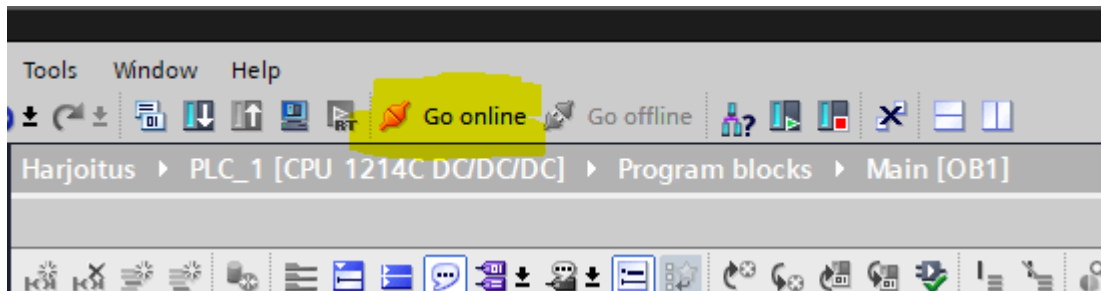
Select **Start below** to be active and press **Finish**



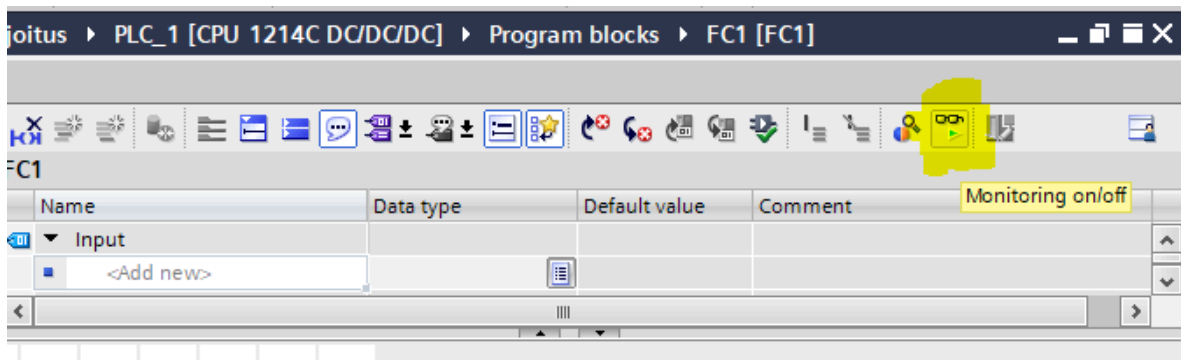
If you receive a notification to select the correct device during the download process, select Type of the PG/PC interface **PN/IE** and PG/PC interface → → **PLCSIM**, press **Start search**. After that, select the device and press **Load**.



Select the desired program (eg. FC1), which you want to simulate and press **Go Online** at the top of the view



The multitasking mode can be turned on by pressing **Monitor's on / off** at the top of the view.



Open the PLCSIM program and select device and **Device configuration** on the left side to display the addresses used in the program.

